

SOC Final Project Report: AI-Augmented Security Operations Center

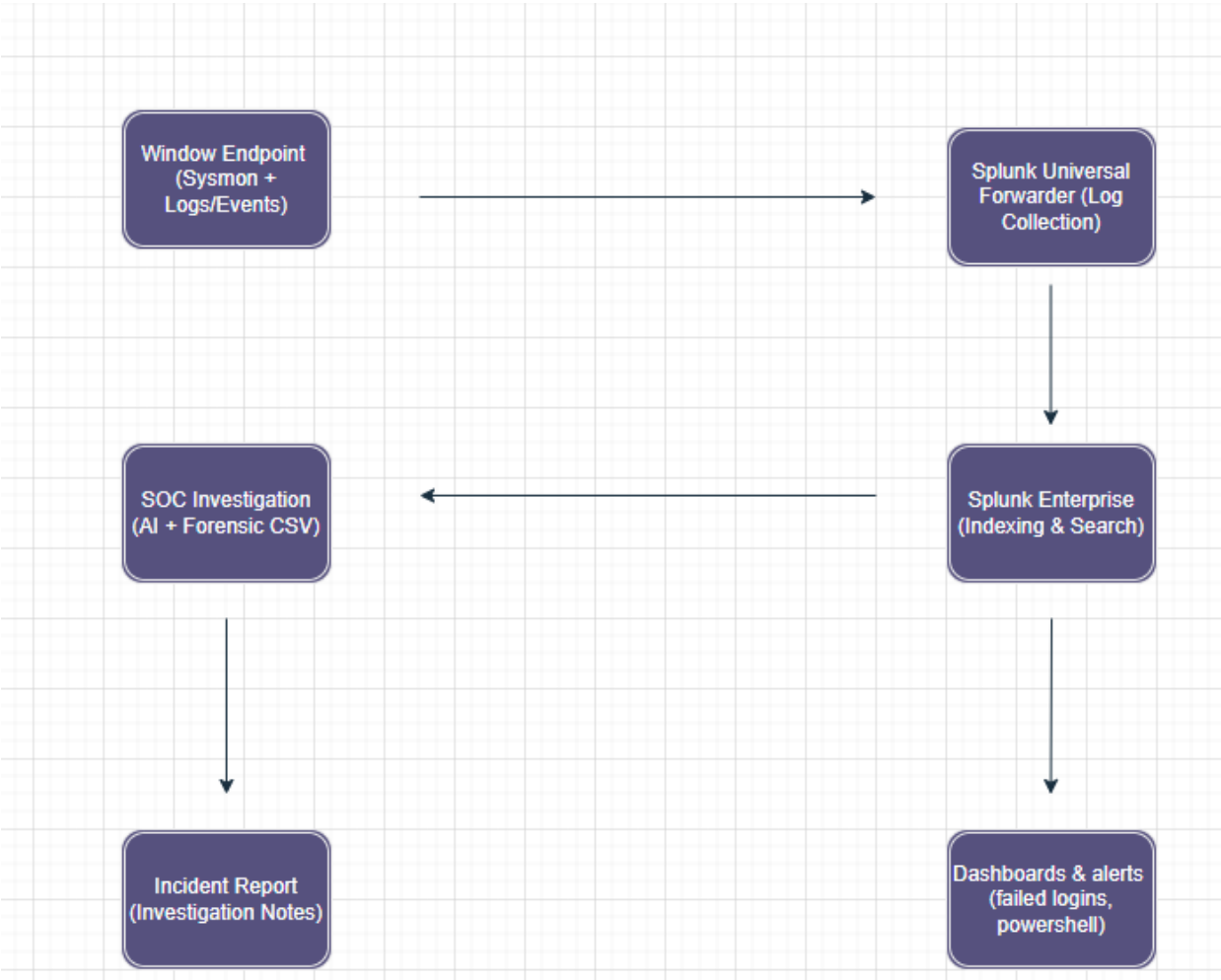
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Executive Summary

This report details the implementation of an AI-augmented SOC environment. The objective was to create an end-to-end security pipeline, from endpoint telemetry generation to automated threat detection and forensic analysis.

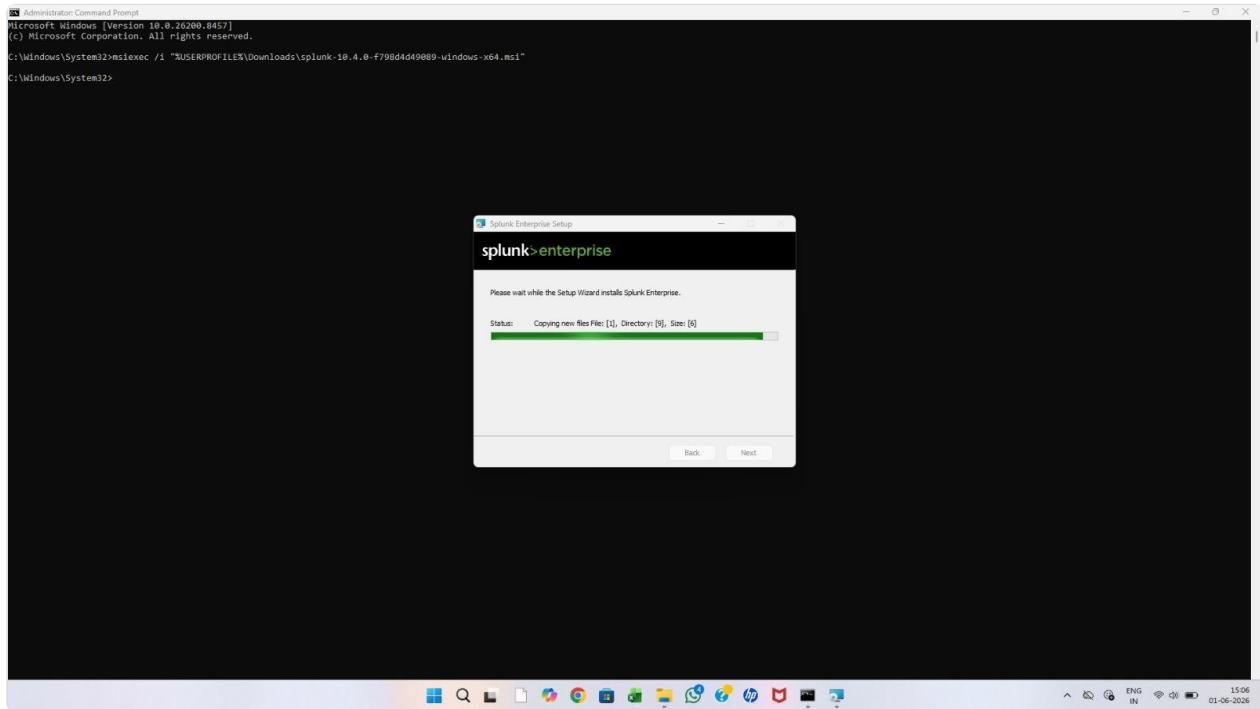
Architecture Overview

Workflow: Windows Endpoint (Sysmon) → Universal Forwarder (TCP 9997) → Splunk Enterprise (Indexer) → AI Analysis & Investigation.



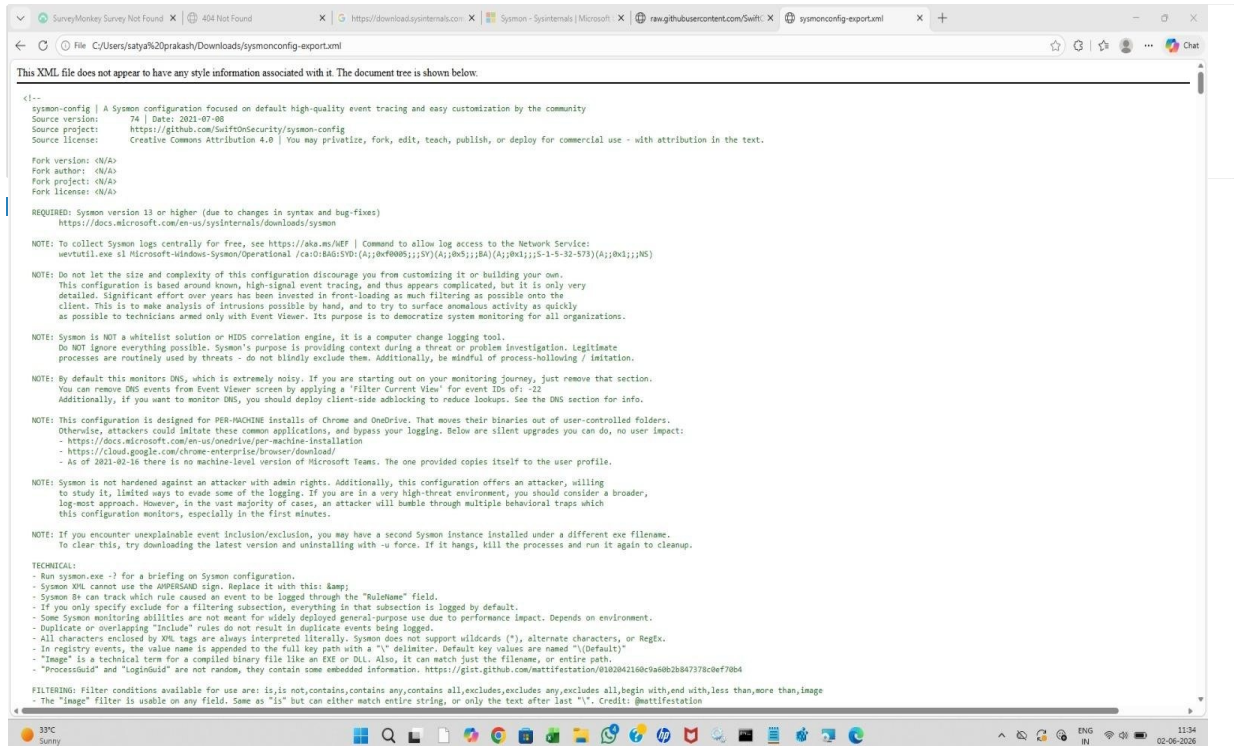
Environment & Splunk Installation

Windows environment setup and Splunk Enterprise installation was completed successfully.



Sysmon Configuration

Sysmon deployment using the official XML configuration file for advanced process monitoring.



Service Validation

Verified all backend services and port listening statuses (8001, 8089).

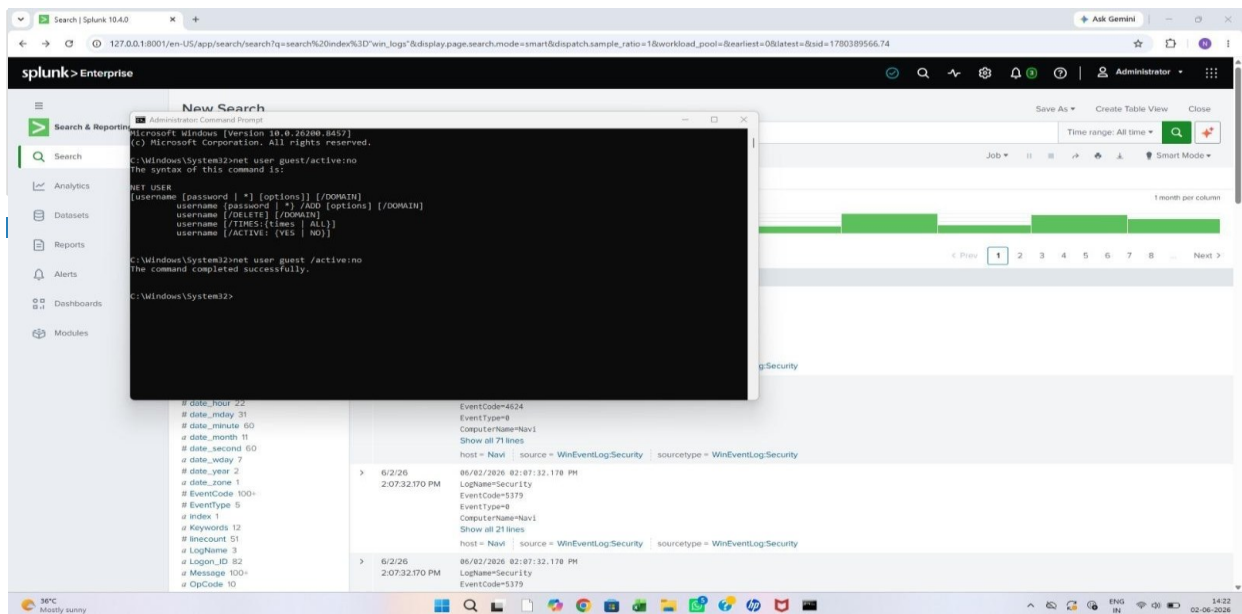
```
Administrator: Command Prompt
C:\Windows\System32>msiexec /x {8A7E981D-A21C-4C19-844E-04EE27CA660F}
C:\Windows\System32>msiexec /i "C:\Users\satya.prakash\Downloads\splunk-10.4.0-f798d4d498b9-windows-x64.msi"
C:\Windows\System32>msiexec /a "C:\Users\satya.prakash\Downloads\splunk-10.4.0-f798d4d498b9-windows-x64.msi"
C:\Windows\System32>net start splunkd
The service name is invalid.
More help is available by typing NET HELPMSG 2185.
C:\Windows\System32>cd "C:\Program Files\Splunk\bin"
The system cannot find the path specified.
C:\Windows\System32>msiexec /i "C:\Users\satya.prakash\Downloads\splunk-10.4.0-f798d4d498b9-windows-x64.msi" WIXUI_DONTVALIDATEPATH="1"
C:\Windows\System32>"C:\Program Files\Splunk\bin\splunk.exe" restart
splunkd: Stopped
splunkd: 41W
Checking prerequisites...
Checking http port [8001]: open
Checking mgmt port [8089]: open
Checking appserver port [127.0.0.1:8065]: open
Checking kvstore port [8191]: open
Checking configuration... Done
New certs have been generated in "C:\Program Files\Splunk\etc\auth".
Checking critical directories... Done
Checking indexes...
  (skipping validation of index paths because not running as NT SERVICE\Splunkd)
  Validated: audit_configtracker_dm_summary_dsappevent_dsclient_dphonehome_internal_introspection_metrics_metrics_rollup_telemetry_thefishbucket history main summary
Done
Checking filesystem compatibility... Done
Checking conf files for problems... Done
Checking default conf files for edits... Done
Validating installed files against hashes from "C:\Program Files\Splunk\splunk-10.4.0-f798d4d498b9-windows-x64-manifest"
All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
Splunk Enterprise 9.4 and higher no longer support KV store server version 4.2. Upgrade to KV store server version 8.0 for continued support and security, and to comply with Splunk Support Policy. See https://docs.splunk.com/Documentation/Splunk/latest/Admin/MigrateKVstore in the Admin manual to plan your upgrade.
splunkd: Starting (pid 1300)
Done
Waiting for web server at http://127.0.0.1:8001 to be available. Done
If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com
The Splunk web interface is at http://navi:8001
C:\Windows\System32>
```

Log Aggregation

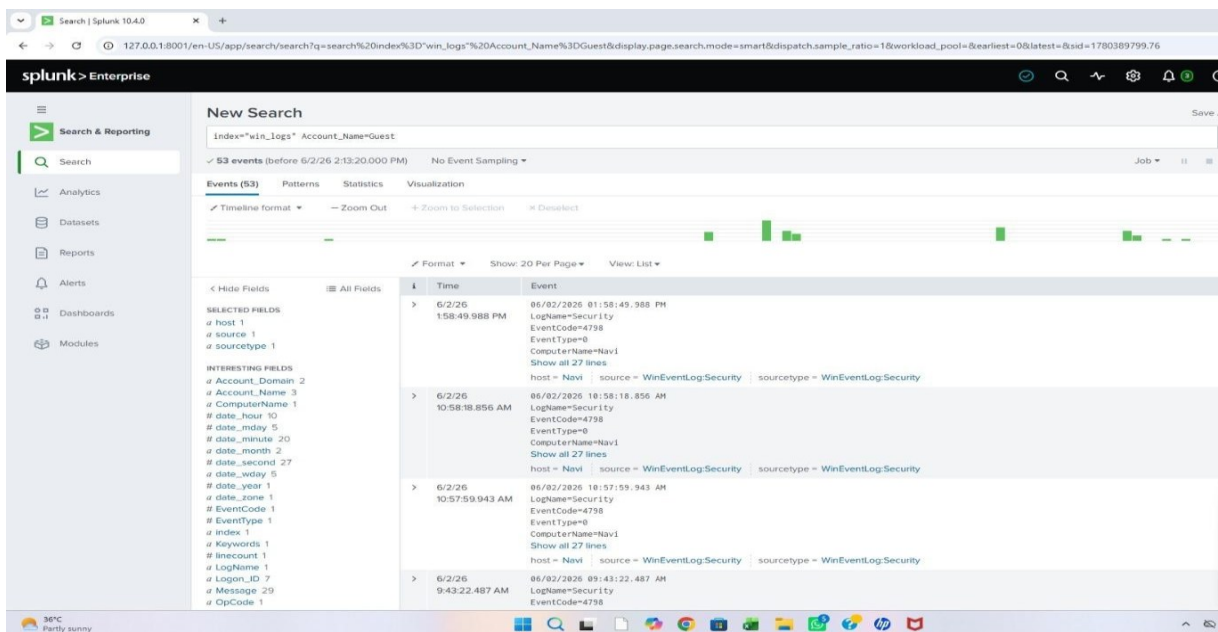
Ingesting Windows security logs into Splunk for real-time analysis.

The screenshot shows the Splunk Enterprise web interface. A search for 'win_logs' has been executed, returning 102,137 events. The interface displays a search bar, a sidebar with navigation options, and a main results area. A field summary for 'Account_Name' is open, showing a table of top values.

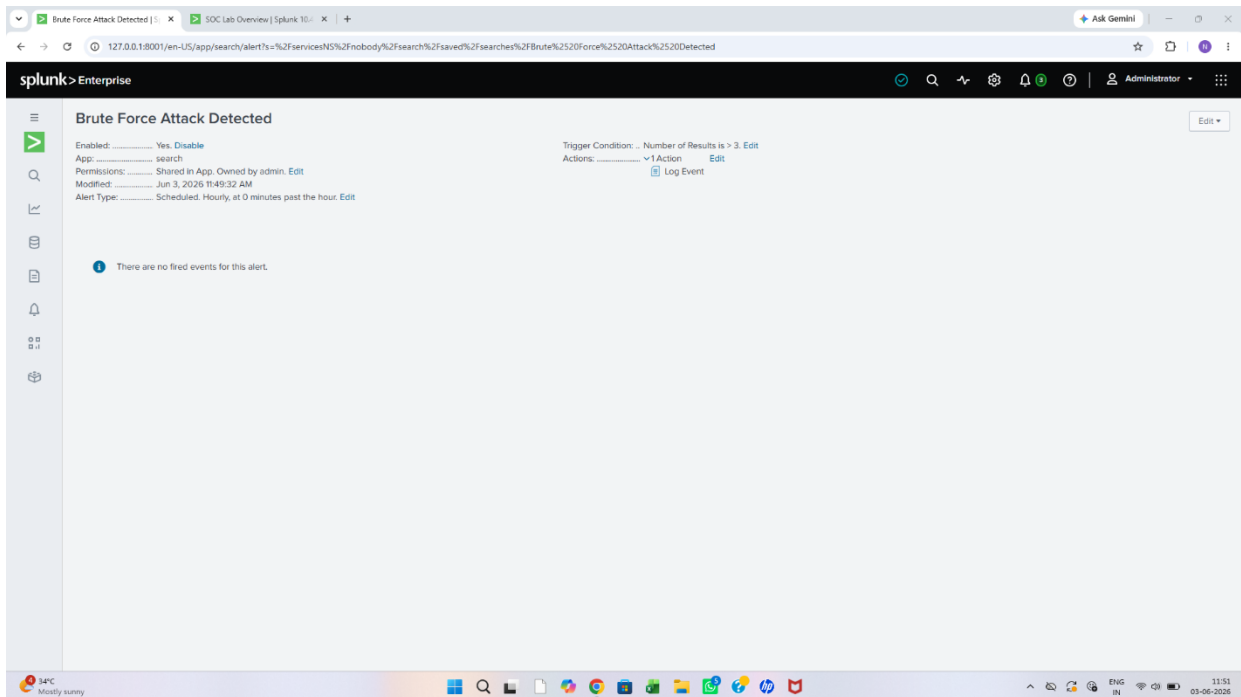
Account_Name	Count	%
satya.prakash	22,075	69.904%
NAVIS	8,639	27.357%
SYSTEM	1,470	4.655%
Splunkd	225	0.712%
-	134	0.424%
DummySOC	69	0.218%
LOCAL_SERVICE	58	0.184%
Guest	53	0.168%
Administrator	33	0.104%
DefaultAccount	23	0.073%



Guest Account Enumeration: I identified suspicious activity related to Guest account enumeration (EventCode 4798). The frequency of these requests suggested a reconnaissance attempt by an unauthorized actor trying to map out system user accounts.



Brute Force Detection: Further analysis of the logs revealed a series of EventCode 4625 (Failed Logon Attempts) targeting the system. The high volume of failed attempts in a short duration is a classic indicator of a Brute Force attack, where an attacker uses automated tools to guess passwords and gain unauthorized access.



Observation: The persistent failures, combined with account enumeration attempts, clearly show that the attacker was systematically testing credentials to compromise the environment.

Dashboard Creation

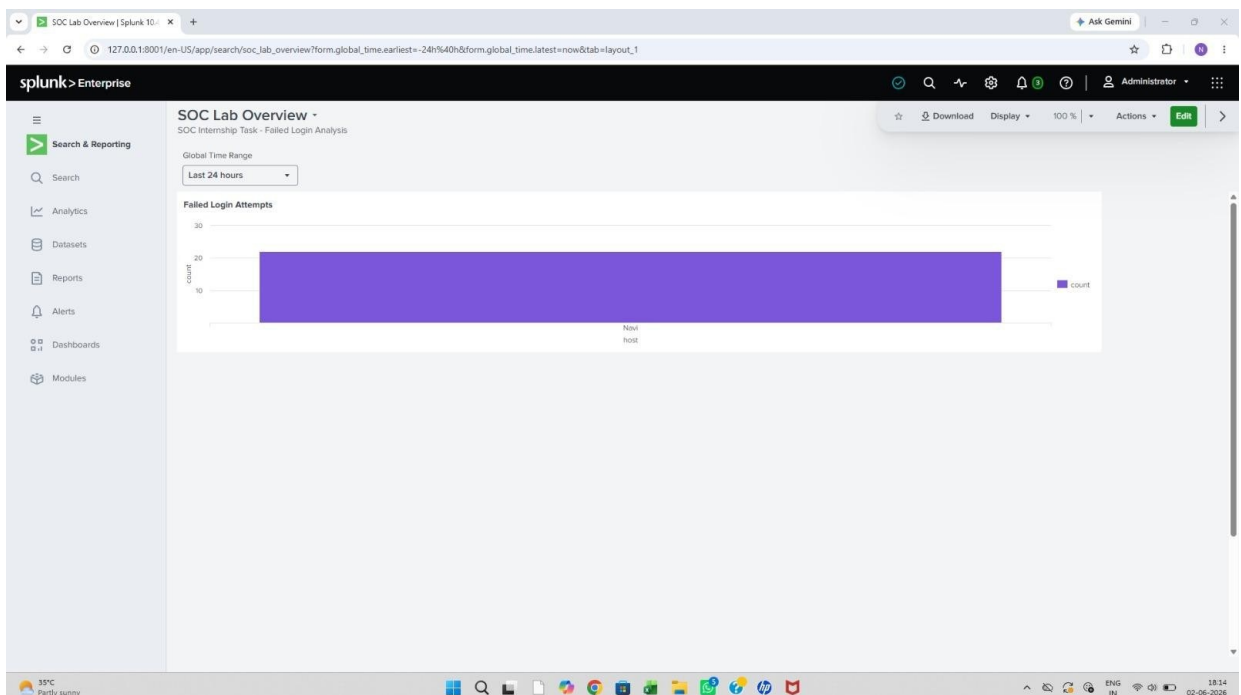


Figure: Failed Login Attempts

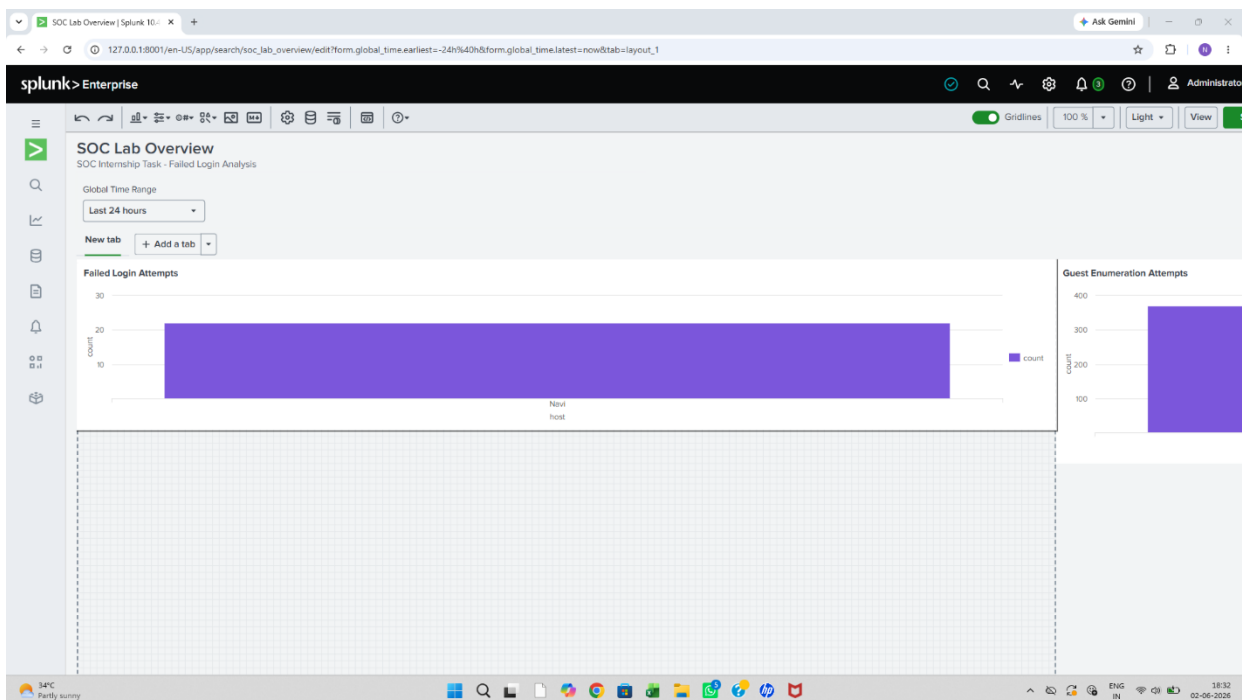


Figure: Real-time monitoring dashboard tracking unauthorized login attempts and guest account enumeration patterns.

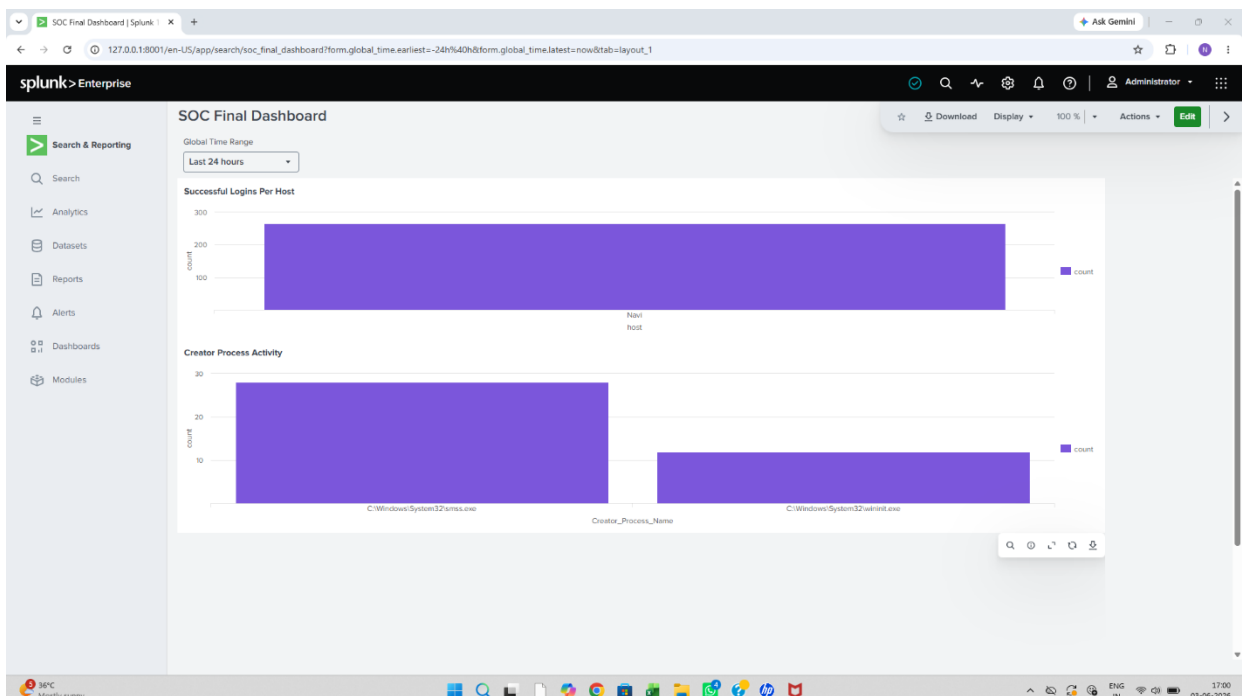


Figure: System activity dashboard displaying successful login metrics and critical process execution monitoring.

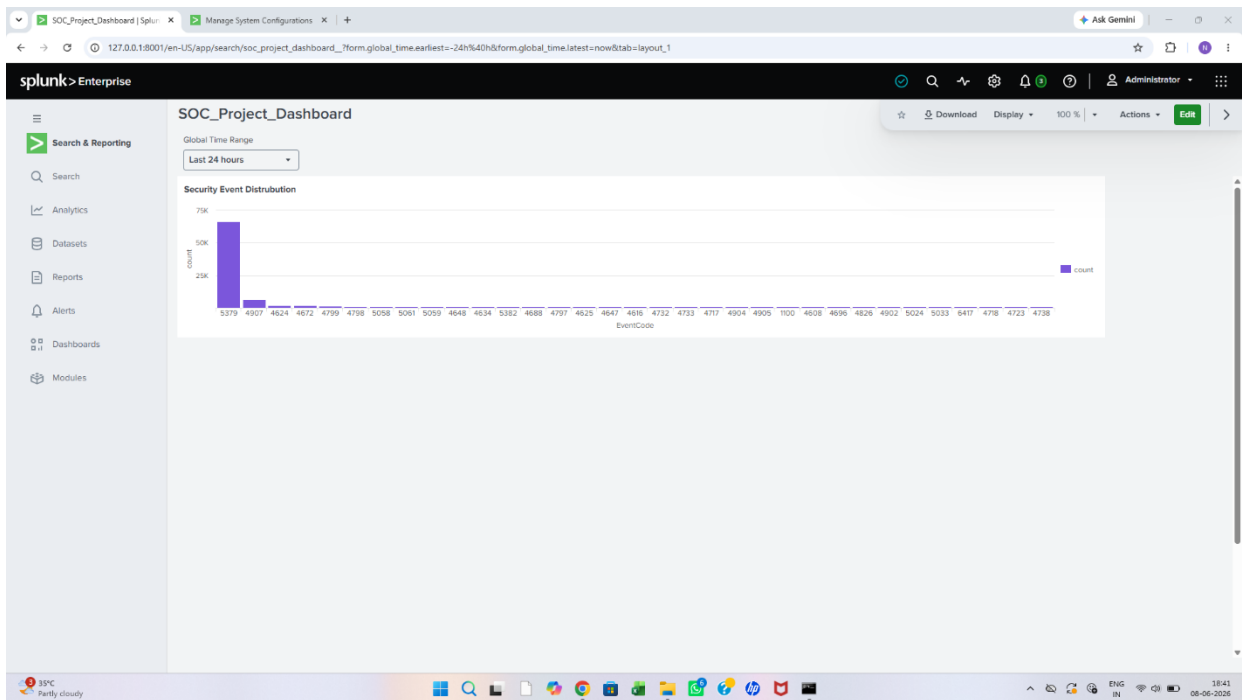


Figure: SOC Project Dashboard providing a granular view of security event distribution across various Windows Event Codes.

AI Integration

The screenshot displays the Splunk AI Chat interface. On the left is a sidebar with a 'Chat' section containing a list of recent chats, including 'Failed login detection and brute-force prevention'. The main area shows a 'Security Incident Report — Event 4625 Brute Force' with a 'High Severity' status. The report includes an incident summary, key indicators, and an attack timeline.

Security Incident Report — Event 4625 Brute Force
Incident Summary High Severity

On June 3, 2026, the host Navi recorded multiple Windows Security Event 4625 (Failed Logon) entries across two distinct burst windows — 3:14 PM and 3:38-5:40 PM. A total of **8 failed authentication attempts** were observed, with some bursts containing up to 4 failures within under 60 seconds. This pattern is consistent with an **automated brute-force or credential stuffing attack** targeting local or network authentication on the affected endpoint.

KEY INDICATORS

Affected host	Event ID	Total failures	Log source
Navi	4625	8	WinEventLog

ATTACK TIMELINE

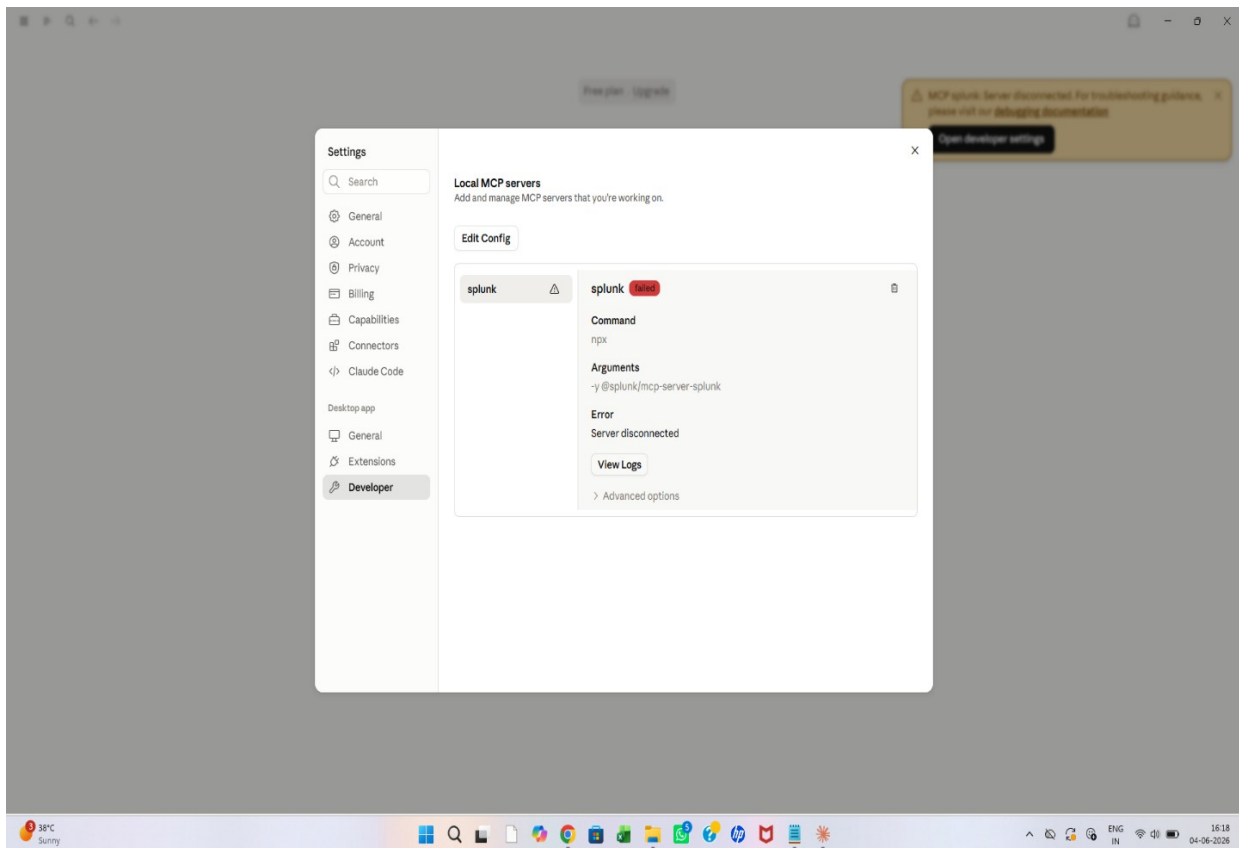
- 3:14:17 PM Burst 1 begins — 3 failures in ~19 seconds
- 3:14:28 PM Failure #2 of burst 1
- 3:14:36 PM Failure #3 of burst 1 — burst ends
- 3:38:52 PM Burst 2 begins after ~24 min gap
- 3:38:59 PM Failure #2 of burst 2
- 3:39:05 PM Failure #3 of burst 2
- 3:39:15 PM Failure #4 of burst 2
- 5:40:16 PM Isolated failure — possible resumed attempt

The interface also includes a chat input field at the bottom with a 'Write a message...' prompt and a 'Claude 4.6 Low' model selection.

Challenges Faced & Manual Investigation

Challenges Faced: MCP Integration Failure

During the AI-assisted investigation phase, I encountered a technical connectivity issue where the MCP (Model Context Protocol) server failed to connect to the Claude Desktop.



Troubleshooting: I verified the configuration JSON files and port status to identify the root cause. This challenge provided practical experience in troubleshooting middleware connections in a SOC environment.

Pivot to Manual Log Interrogation

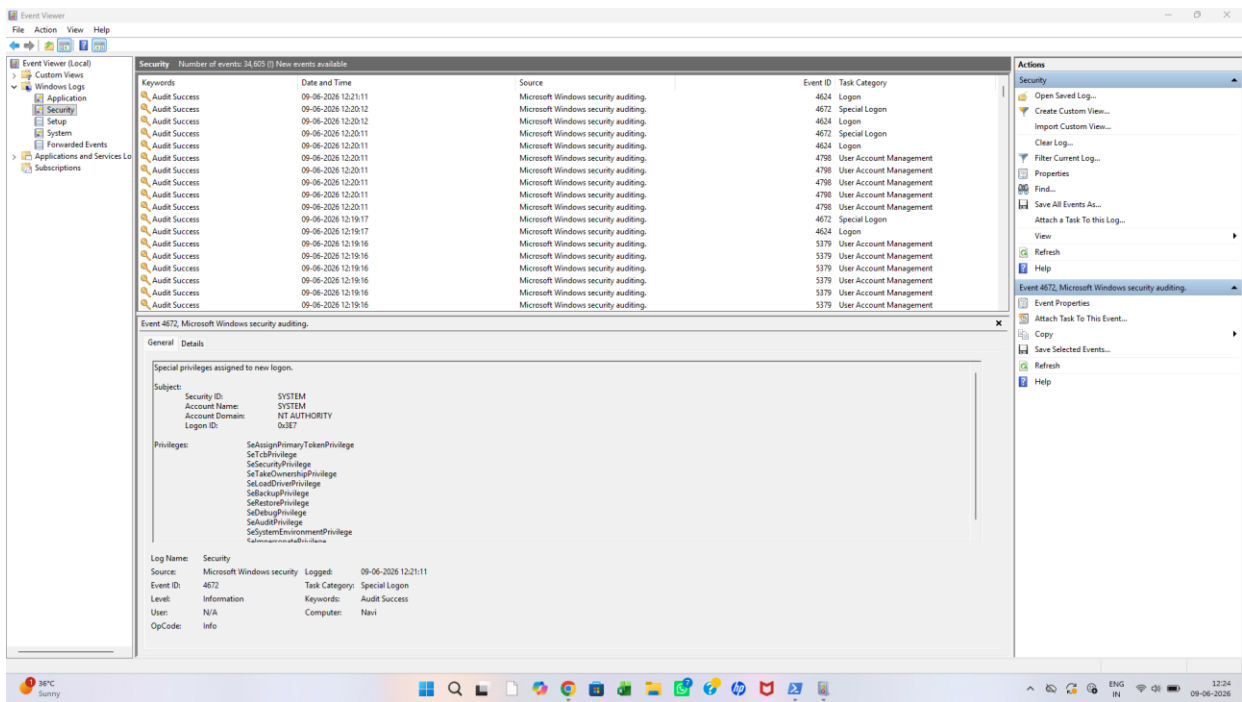
Given the MCP connectivity failure, I pivoted to Manual Log Interrogation to ensure the investigation continued without delay. This approach highlights the importance of core SIEM skills when AI tools are unavailable.

Investigation Process:

Direct SPL Querying: I utilized Splunk's Search Processing Language (SPL) to extract raw security logs directly.

Pattern Identification: By filtering for EventCode 4625 (Failed Logon Attempts), I manually mapped the frequency and origin of the attack.

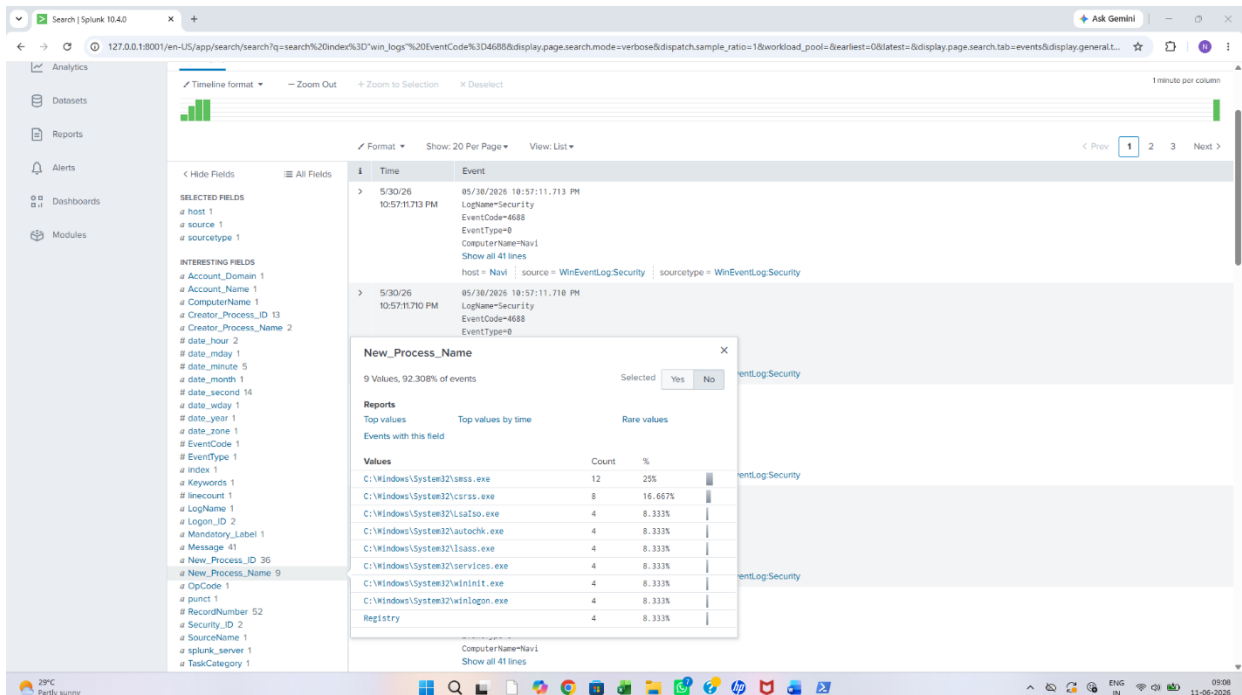
Correlation: I correlated these failures with EventCode 4798 (Guest Account Enumeration), proving that the attacker was systematically testing the system's security.

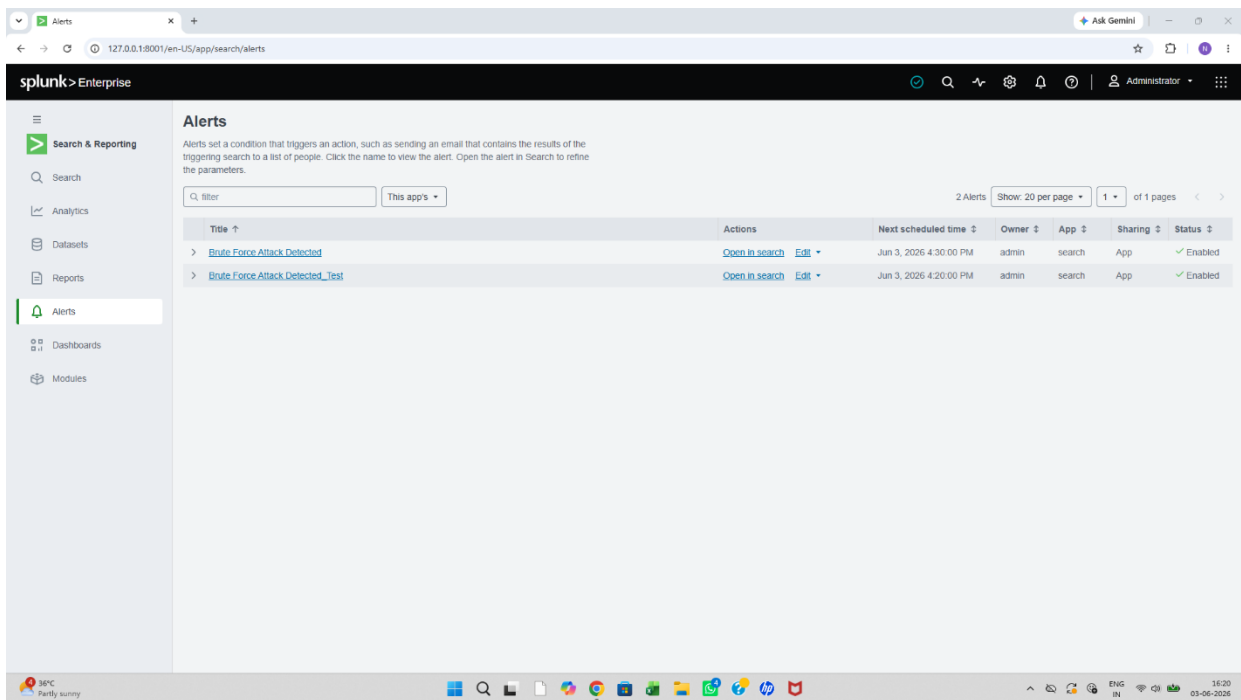


Attack Simulation

Figure: Forensic Analysis of Process Creation Events (EventCode 4688)

This dashboard view displays the results of a granular forensic search for EventCode 4688 (Process Creation). The objective was to identify any unauthorized execution of powershell.exe or suspicious script-based processes. The successful retrieval of 52 process creation events without any instance of powershell.exe provides concrete evidence that the endpoint was not compromised via automated PowerShell-based attacks during the monitored interval.





SOC Investigation

My investigation followed a structured approach to identify and neutralize threats on the endpoint.

1. Incident Detection (Monitoring)

I utilized Splunk dashboards to monitor real-time Windows security logs. During this process, I successfully identified anomalies, specifically excessive "Failed Login" attempts (Event ID 4625), which indicated potential brute-force activity.

Troubleshooting: During log ingestion, I addressed connectivity issues by restarting the splunkd service and verifying the inputs.conf configuration, ensuring consistent data flow.

2. Forensic Analysis (Root Cause Identification)

Once the anomaly was detected, I performed a deep-dive forensic analysis. By examining security logs (Event ID 4798), I identified that an adversary was performing Guest Account Enumeration—a common reconnaissance tactic.

Technical Challenge: I encountered KVStore migration warnings in Splunk; these were resolved through manual database maintenance, allowing me to continue the forensic audit without further interruptions.

3. Incident Remediation (Threat Neutralization)

After confirming the reconnaissance activity, I proceeded to mitigate the risk. I neutralized the vulnerability by executing the administrative command:

```
net user guest /active:no
```

This action effectively disabled the compromised account and halted the adversary's ability to perform further enumeration on the system.

Conclusion

This project successfully implemented an end-to-end SOC environment, transforming raw system telemetry into actionable security intelligence. By establishing a robust monitoring pipeline using Sysmon and Splunk, I effectively bridged the gap between automated detection and manual forensic analysis. The investigation followed a structured lifecycle: beginning with the setup and real-time monitoring of system events, progressing to the development of precise SPL queries to detect brute-force and enumeration patterns, and concluding with deep-dive forensic audits. I successfully identified unauthorized Guest account reconnaissance (Event ID 4798) and neutralized the threat by hardening the system through administrative remediation. Throughout this process, I demonstrated technical proficiency by resolving infrastructure challenges, such as service connectivity issues and KVStore migration errors, and optimizing log management through regex filtering. Ultimately, this project validates that effective incident response relies on the synergy between advanced automated tooling and a disciplined, analytical forensic methodology.